VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Local Function MPU

AFLCMC/WA

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1.0 Overview

A Local Function is a Level 1 Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). Local Functions provide computing for the Multi-Function Aperture (MFA) Layer and allow for low-latency processing of data generated by an MFA MPU. Even though Local Function is logically part of the MFA Layer, the processing element used by the Local Function MPU may physically exist in the MFA or a Digital Payload Processor (DPP).

1.1 Purpose

The Local Function MPU Volume introduces the Local Function MPU and describes its key features, interfaces, and compliance requirements. The key features provide design guidance on what is typically expected for the Local Function MPU. The interface diagram contains a graphical representation of the types of interfaces that can exist (from the perspective of the Local Function MPU). **Table 2.3.1** lists and describes the Local Function MPU interfaces. Compliance requirements describe what is required for the Local Function MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The Local Function Volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. AnAMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the Local Function Volume.

Standard	Title	Version/Date
AMS GRA Models	AMS GRA Architecture Model	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	IR MEL IDD	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	MFA Local Function Contract Template	14.0

AMS GRA Volumes	RF MEL IDD	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Local Function.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Local Function Volume.

Standard	Title	Version/Date
AMS GRA Volumes	MFA MPU Volume	14.0
AMS GRA Volumes	Software Architecture Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Local Function and Related Entities.

2.0 Local Function

A Local Function enables the business and technical goals of AMS GRA by reducing vendor lock, decoupling components, and providing Open interfaces—which in turn allows for more competition. This construct uses the available processing power in the hardware to enable other vendors to create improved algorithms that can meet the timing and data rates necessary by interacting more directly with the MFA. Local Functions increase capability by leveraging lower-latency and/or higher-bandwidth access to MFA resources through MFA Compute as compared to Multi-Function Processor (MFP) software Services.

2.1 Key Features

Table 2.1.1 is not prescriptive but identifies a number of the features and attributes that AMS GRA interfaces are designed to enable.

Key Feature	Benefit
Proximity to MFA Hardware	Provides a way to handle requirements not met by using a Service (e.g., timing, data rate, etc.).
Third Party Insertion	Allows third parties to implement low-latency functions.

Table 2.1.1: The Local Function Key Features Table Describes the Specific Functionality Supported by a Local Function Instantiation.

2.2 Interface Diagram

Figure 2.2.1 depicts the Local Function and its execution environment provided by the MFA MPU logical interfaces. Interface connections denoted as pass-through only represent situations where the sending MPU is not the original source and does not use or modify what is provided to the receiving MPU.

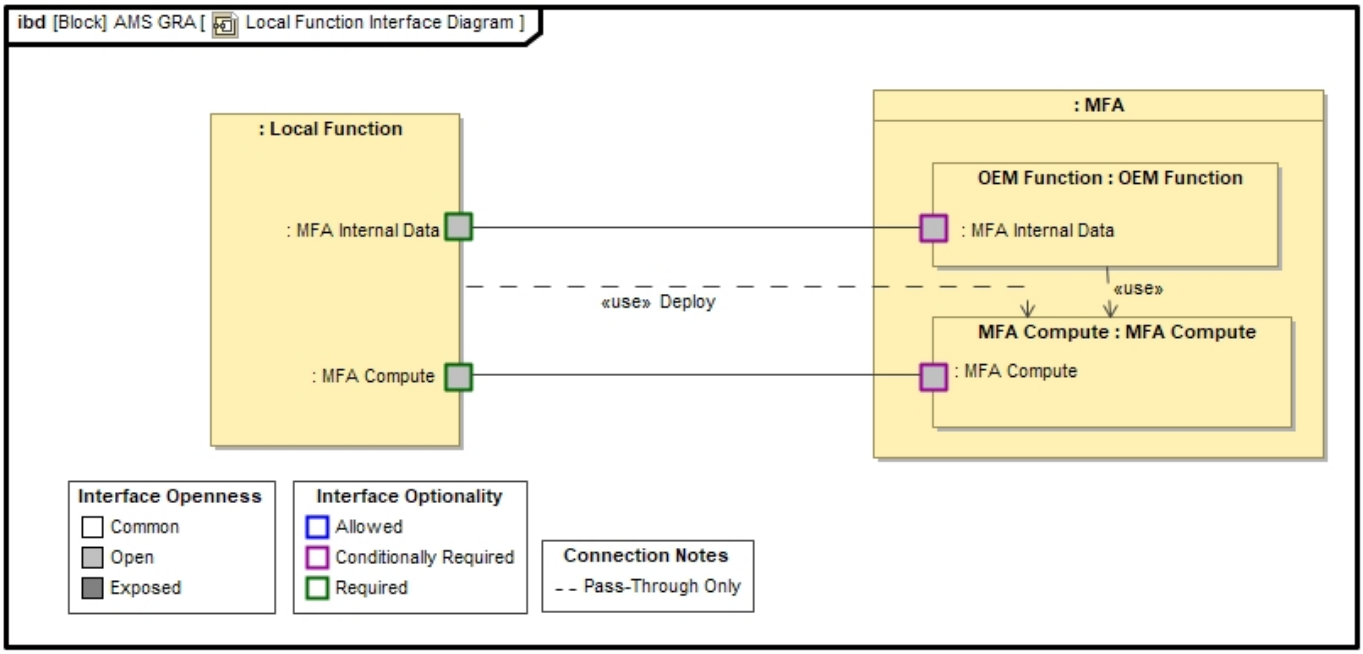


Figure 2.2.1: This Local Function Interface Diagram Shows Logical Interfaces of the Local Function and Their Connections with Other MPUs.

As depicted in the interface diagram in the MFA Volume and in this interface diagram, the Local Function interacts with the MFA. In particular, the Local Function interacts with the Original Equipment Manufacturer (OEM) functions for access to MFA Data and internal functions. Also, the Local Function interacts with the MFA Compute resources—which may or may not be physically part of the MFA. In the case where a DPP in the Compute Cluster Farm (CCF) realizes the MFA Compute, the Compute Interface becomes the DPP Application Programming Interface (API).

Services only communicate with Local Functions through MFA Encapsulation Layer (MEL) constructs. Refer to appropriate MEL Interface Description Documents (IDDs) for implementation details.

2.3 Interfaces Table

Table 2.3.1 identifies the Local Function interfaces depicted in **Figure 2.2.1**. When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally Required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

Interface	Documentation	Interface Optionality
MFA Compute	The MFA Compute Interface allows OEM Functions and Local Functions to interact with the processing resources of the MFA Compute. When the MFA Compute is instantiated within the physical MFA, the Compute Interface is an Open interface defined by the MFA provider. When the MFA Compute is instantiated within a DPP dedicated to the MFA, the Compute Interface is the DPP API.	Required
MFA Internal Data	The MFA Internal Data interface is the set of data interfaces provided by the MFA OEM for Local Functions to utilize. These Data interfaces are defined by the Local Function DK delivered with the MFA.	Required

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 states the applicable MPU Compliance Requirements for the Local Function MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

See the Mission System Volume for additional details about MPU Fault Reporting.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-023	An MPU shall report status to the MFA IAW the DK.	Tier 1	Demonstration	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-039	An MPU shall provide a CLASSIFICATION.md file defining security classification of included file contents.	Tier 2	Inspection	
GRA-MPU-040	An MPU shall provide a DATARIGHTS.md file defining data rights of included file contents.	Tier 2	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-051	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 1	Inspection	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUs and CDS MPUs report faults as defined in the CDS and CU IDD. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.

Table 3.0.1: A Local Function Volume's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

AMS	Agile Mission Suite
API	Application Programming Interface
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DAST	Dynamic Application Security Testing
DK	Development Kit
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
RoT	Root of Trust
SAST	Static Application Security Testing
SK	Starter Kit
UCI	Universal Command and Control Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24, 2025	SK13.0
B	This revision did not introduce any significant prose changes. Updates GRA-MPU-051 for clarity.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Local Function Volume Between Documentation Releases.